ARCnordic Tid

Kodeoversigt og funktioner

Arkitektur, GRASP og OOP, moduler og funktioner, datamodel, sikkerhed, test og vedligeholdelse

24.083	396	208	106
linjer kildekode	dokumenterede funktioner/typer	automatiske tests	API-endpoints

Udarbejdet 9. november 2026 · version 547a584 (31 commits) · github.com/Deb2121/ARC-Time
Genereres fra koden med `node scripts/code-report.mjs` — inventarlisterne passer altid med koden.

Indhold

1. Resumé — hvad systemet er, og hvor det står
2. Arkitektur — lag, pakker og en request fra start til slut
3. GRASP og OOP i koden
4. Clean code — konventioner der holder koden vedligeholdelig
5. Sikkerhed og adgangsstyring (rollematrix)
6. Data og privatliv (GDPR)
7. Test og kvalitetssikring
8. Vedligeholdelse — sådan tilføjer man en funktion
9. Moduler og funktioner (inventar)
10. API-endpoints
11. Datamodel (tabeller)
12. Beslutningslog
13. Kendte begrænsninger og anbefalinger

1. Resumé

ARCnordic Tid er et PSA-system (tid, projekter, budget, godkendelse, fakturering, udlæg, fravær og flex, ressourceplanlægning, løneksport og rapportering) bygget som en ren genimplementering. Systemet er bygget i faser: M0 fundament, M1 tidsregistrering og godkendelse, M2 fakturering og perioder, M3 drift (udlæg, ressourceplan, dashboard, Outlook, analyse), M4 løneksport og import, samt rettelser efter kodegennemgang (adgangspolitik, privatliv, revisionsspor, GDPR, backup).

1.900 packages/domain · 35 filer	3.851 packages/db · 15 filer	297 packages/integrations · 6 filer	130 packages/ui · 1 filer
7.770 apps/api · 57 filer	9.630 apps/web · 45 filer	505 apps/analytics · 9 filer	

Teknologi: TypeScript overalt (Node 26, pnpm-monorepo), Next.js 16 og React 19 til web, Fastify 5 og Zod til API, Drizzle ORM og PostgreSQL 17 med row-level security, Python 3.12 og FastAPI til analyse. Biome (lint og format), Vitest, pytest og Playwright til test. Alt kører lokalt i Docker (D-017); driftsforslag ligger i `docs/operations.md`.

Designprincipper: forretningsregler er rene funktioner i domænelaget; alle eksterne systemer sidder bag adaptere og er slået fra som standard; al adgangslogik bor ét sted (`AccessPolicy`); hver fil har en forklarende header og hver eksporteret funktion en JSDoc/docstring, håndhævet automatisk.

2. Arkitektur

Koden er et monorepo med tre apps og fire pakker. Afhængigheder går kun nedad i diagrammet: domænet kender ingen andre lag, og web-appen taler kun med API'et via HTTP.

apps/web — Next.js (brugerflade)
Sider pr. skærm (`app/(app)/…`), genbrugelige komponenter, i18n (dansk/engelsk). Viser kun handlinger, som serverens rollematrix tillader (kapabiliteter fra `/me`).

HTTP + session-cookie + CSRF-header

apps/api — Fastify (GRASP Controller + services)
`routes/`: validér (Zod) → tjek adgang (`requireCapability`) → kald service → svar. `services/`: arbejdsgange (godkendelse, fakturaer, løn, GDPR …). Baggrundsjob: outbox, påmindelser, opbevaringsfrist.

funktionskald

packages/domain — forretningsregler (Information Expert)
Rene funktioner: normtid, flex, ferie, satser, faktureringsforslag, nøgletal, løn, import, GDPR og `AccessPolicy`. Ingen I/O → hurtige, sikre tests.

packages/db — Drizzle + PostgreSQL
`schema.ts` (alle tabeller), migreringer, RLS pr. virksomhed via `withTenant()`, audit-skriver, transaktionelle arbejdsgange (bogføring, perioder).

packages/integrations — adaptere (Polymorphism / Protected Variations)
e-conomic (regnskab) og Microsoft Graph (kalender). Interfaces gør dem udskiftelige; slået fra uden konfiguration.

apps/analytics — Python FastAPI (skrivebeskyttet)
Nøgletal, Excel og Power BI-datasæt via en databaserolle med kun SELECT. Formlerne spejler `kpi.ts` og testes med samme eksempler.

En request fra start til slut — "gem en celle i ugesedlen"

1. Brugeren skriver 7,5 i en celle → `week-grid.tsx` sender `PUT /api/v1/timesheet/entries` med CSRF-header.
2. `auth/plugin.ts` slår sessionen op (kun hash gemmes), indlæser roller og ekstra rettigheder og afviser manglende CSRF.
3. `routes/timesheet.ts` validerer body med Zod og kalder `saveCell()` i `services/timesheet.ts`.
4. Servicen henter data i en transaktion med `app.tenant_id` sat (RLS), og kalder domænets `validateEntry()`: 24-timers-grænse, 0,25-trin, lukket periode, påkrævet kommentar.
5. Gemmer rækken; projektleder-stemplet nulstilles, hvis rækken ændres. Fejl kastes som `AppError` og oversættes centralt til `{ error: kode }`, som web-appen viser på dansk.

3. GRASP og OOP i koden

Princip	Hvor det ses
Information Expert	Beregninger bor hos de data, de kræver: `norm.ts`, `flex.ts`, `rates.ts`, `invoicing.ts`, `payroll.ts`, `kpi.ts`; `AccessPolicy` ved, hvem der må hvad; `people.ts` ved, hvem der er leder for hvem.
Creator	Den, der ejer noget, opretter det: en uge opretter sin ugeseddel-periode (`ensurePeriod`), et projekt opretter sin første kontrakt og PL-medlemskab, en faktura sine linjer og kildereferencer.
Controller	Kun rutefilerne: validér → autorisér → delegér → svar. Ingen SQL-arbejdsgange eller regler i ruterne.
Low Coupling	Services kaster `AppError` og kender ikke HTTP. Domænet har ingen afhængigheder. Web kender kun API'ets JSON. Eksterne systemer nås kun via adaptere.
High Cohesion	Ét ansvar pr. modul: `project-setup.ts` ændrer projekter, `projects.ts` læser dem; `audit-log.ts` læser sporet, `audit.ts` skriver det.
Polymorphism	`AccountingAdapter`, `CalendarSource`, `CalendarProvider`, `Mailer` — flere implementeringer bag samme interface (rigtige og falske i tests).
Pure Fabrication	Klasser/moduler for god struktur: outbox, audit-skriver, rapport/CSV-bygger, løneksport, GDPR-service, Excel-bygger.
Indirection	`withTenant()` mellem kode og database; transaktionel outbox mellem bogføring og regnskabssystem; `/me`-kapabiliteter mellem server-regler og knapper i UI.
Protected Variations	RLS, triggers der gør bogførte fakturaer uforanderlige, append-only audit_log, adapter-interfaces, i18n-nøgler, rollematrix som data.

Objektorienteret, hvor det giver mening

Kernen er bevidst **funktionel** (rene funktioner over simple data er lettest at teste og at spejle i Python). Objekter bruges, hvor de modellerer noget med tilstand og adfærd, eller en udskiftelig samarbejdspartner:

Klasse / interface	Hvorfor et objekt
`AccessPolicy`	Indkapsler én persons roller + rettigheder. Metoderne `can()`, `capabilities()`, `seesAbsenceReason()`, `canReadAudit()`. Matrixen er privat data — kaldere ser aldrig rollelister.
`EconomicAdapter`, `GraphCalendarSource`	Implementerer interfaces (polymorfi); kan udskiftes uden at røre kaldere; tests bruger falske udgaver.
`AppError, InvoiceError, PeriodError, IntegrationError`	Fejltyper bærer HTTP-status/"prøv igen"-flag; én central fejlhåndtering oversætter dem.
`Caller (Python dataclass)`	Uforanderligt værdiobjekt for den indloggede i analysetjenesten.

Eksempel — hele rollematrixen spørges gennem én metode:

```
/** True when the matrix allows the action for this person. */
  can(capability: Capability): boolean {
    const rule = MATRIX[capability];
    if (hasRole(this.roles, ...rule.roles)) return true;
    return Boolean(
      rule.right && this.rights.includes(rule.right.right) && hasRole(this.roles, ...rule.right.rightFor),
    );
  }
```

4. Clean code — konventioner

- **Én sandhed pr. regel.** Rollematrix (`access.ts`), beløbslinjer (`isAmountLine`), nøgletal (`kpi.ts` + Python-spejl testet med samme eksempler).
- **Enheder er heltal.** Penge i øre, timer i centitimer (7,5 t = 750), procent i basispoint (81,08 % = 8108). Ingen kommatal i beregninger → ingen afrundingsfejl. Datoer er lokale ISO-datoer i Europe/Copenhagen.
- **Dokumentation håndhæves.** `scripts/check-docs.mjs` kræver header i hver fil og JSDoc/docstring på hver eksport (TypeScript og Python) — del af `pnpm check`, pre-commit-hook og CI.
- **Navne fortæller hvad.** Funktioner er verber (`saveCell`, `exportPayroll`), regler er navneord (`retentionCutoff`). Fejlkoder er maskinlæsbare (`period_closed`) og oversættes i UI.
- **Små, tynde lag.** Ruter er korte; arbejdsgange i services; regler i domænet. Ingen forretningslogik i React-komponenter ud over visning.
- **Sikker som standard.** Integrationer, mail og kalender er slået fra uden konfiguration; dev-login kan aldrig være slået til i produktion.
- **Automatisk formattering og lint** (Biome) — ingen stilkampe i code review.

5. Sikkerhed og adgangsstyring

- **Login:** uigennemsigtige session-tokens i httpOnly-cookie (kun SHA-256-hash gemmes), CSRF-token på alle ændringer, rate limit på login. Microsoft Entra ID (OIDC + PKCE) er bygget, men slået fra lokalt.
- **Virksomhedsadskillelse:** PostgreSQL row-level security med en ikke-ejer app-rolle; hver forespørgsel kører med `app.tenant_id` sat. Analysetjenesten har sin egen rolle med kun SELECT.
- **Uforanderlighed:** bogførte fakturaer er låst af database-triggers; revisionssporet (`audit_log`) kan kun indsættes i — app-rollen har ikke UPDATE/DELETE.
- **Funktionsadskillelse (D-025):** Admin konfigurerer, Økonomi bogfører, krediterer, lukker perioder og eksporterer løn.
- **Privatliv (OQ-21):** sygdom vises som "Fravær" uden kommentar for alle andre end medarbejderen, lederen og Økonomi — i ugevisning, rapporter og analyse.

Rollematrix (genereret fra `access.ts`)

Handling	Medarbejder	Projektleder	Leder	Økonomi	Ledelse	Admin
`project.create`	–	◐	–	✓	–	✓
`invoice.prepare`	–	✓	–	✓	–	–
`invoice.prepare_all`	–	–	–	✓	–	–
`invoice.book`	–	–	–	✓	–	–
`invoice.credit`	–	–	–	✓	–	–
`period.view`	–	–	–	✓	✓	✓
`period.close`	–	–	–	✓	–	–
`payroll.export`	–	–	–	✓	–	–
`payroll.setup`	–	–	–	✓	–	✓
`expense.approve_any`	–	–	–	✓	–	–
`timesheet.approve_unmanaged`	–	–	–	✓	–	–
`audit.read`	–	–	–	✓	✓	✓
`audit.read_all`	–	–	–	–	✓	✓
`gdpr.manage`	–	–	–	–	–	✓
`settings.manage`	–	–	–	–	–	✓
`settings.read`	–	–	–	✓	–	✓
`pricelist.manage`	–	–	–	✓	–	✓
`people.import`	–	–	–	–	–	✓
`data.import`	–	–	–	✓	–	✓
`kpi.company`	–	–	–	✓	✓	✓
`analytics.token`	–	–	–	✓	✓	✓

◐ = kun med ekstra rettighed (fx `create_projects`, tildeles under Medarbejdere). Hvem der ses (data-omfang: egne, egne medarbejdere, egne projekter) afgøres i `services/people.ts` og `services/projects.ts`.

6. Data og privatliv (GDPR)

- **Dataudtræk:** Admin → Medarbejdere → "Hent alle persondata" giver én JSON-fil med alt om personen og intet om andre (`exportPersonalData`).
- **Sletning af fratrådte:** "Slet personoplysninger" pseudonymiserer navn, e-mail, initialer og lønnummer, fjerner kommentarer og sletter ikke-bogføringsdata. Bogføringsdata bevares i 5 år efter regnskabsåret. Blokeres, hvis personen er aktiv, projektleder på aktive projekter eller leder for andre (`erasePerson`, `erasureBlockers`).
- **Deaktivering** logger ud med det samme (sessioner og kalender-tokens slettes).
- **Natligt job** sletter udløbne sessioner/tokens og logs ældre end 13 måneder og rapporterer bogføringsdata ud over fristen (sletning af regnskabsdata er en bevidst handling efter aftale med revisor).
- **Fortegnelse:** `docs/gdpr/processing-inventory.md`.

Drift og backup

Lokalt tages en backup dagligt ved opstart (14 gemmes) og ved nulstilling; gendannelse er verificeret. Forslag til produktion (Azure EU, administreret PostgreSQL med point-in-time restore, Key Vault, Blob Storage, Microsoft 365-mail) og flytning ud af OneDrive står i `docs/operations.md`.

7. Test og kvalitetssikring

Område	Filer	Tests
Domæne	11	87
Database / RLS	1	5
Integrationer	2	6
API (integration mod rigtig database)	12	77
Python-analyse	3	13
E2E i browser (Playwright)	12	20
I alt		**208**

- **Domænetests** bruger de regneeksempler, der står i specifikationen (fx flex +0,5; 49 % færdig → 117.600 kr; 3,00 feriedage).
- **API-tests** kører mod den rigtige database med RLS. Destruktive forløb (låsning af lønperioder, import, GDPR-sletning) kører i en transaktion, der rulles tilbage — demodata røres ikke.
- **E2E-tests** kører i en rigtig browser på dansk mod produktionsbygningen i CI.
- **Kvalitetsport:** `pnpm check` = lint + dokumentationstjek + typecheck + alle tests. Pre-commit-hook kører lint, docs og typecheck; GitHub Actions kører det hele inkl. migrering, drift-tjek og E2E.

8. Vedligeholdelse — sådan tilføjer man en funktion

1. **Regel først:** skriv beregningen som ren funktion i `packages/domain/src/<emne>.ts` med JSDoc og et eksempel, og en test med tal fra specifikationen.
2. **Data:** tilføj tabel/kolonne i `packages/db/src/schema.ts` → `drizzle-kit generate`; nye tabeller med `tenant_id` får RLS via skabelon-migreringen (RLS-testen fejler ellers).
3. **Arbejdsgang:** service-funktion i `apps/api/src/services/`; skriv til revisionssporet ved ændringer af økonomiske data.
4. **Adgang:** tilføj en kapabilitet i `AccessPolicy`, hvis handlingen er ny; brug `requireCapability` i ruten og `can(me, …)` i web.
5. **Rute:** tynd Controller med Zod-skema i `apps/api/src/routes/`; registrér i `app.ts`.
6. **UI:** side under `apps/web/src/app/(app)/`, tekster i `i18n/da.ts` og `en.ts`.
7. **Test og dokumentation:** API-test (rul tilbage, hvis den skriver), evt. E2E; beslutning i `docs/decisions.md`; kør `pnpm check`; generér denne rapport igen.

Hvor finder jeg …? Arkitekturguiden `docs/architecture.md` (kodekort, GRASP, enheder), specifikationen `docs/spec/*.md`, beslutninger `docs/decisions.md`, åbne spørgsmål `docs/open-questions.md`, drift `docs/operations.md`, demo-gennemgange `docs/demo/`.

9. Moduler og funktioner

Udtrukket automatisk fra kildekoden: filens formål (header) og hver eksporteret funktion/type med første sætning af dens dokumentation. Sidefiler (Next.js-sider) er ikke med — deres formål er deres skærm. Beskrivelserne er på engelsk, fordi kodens kommentarer bevidst skrives på engelsk, så enhver udvikler kan læse dem; brugerfladen er på dansk og engelsk.

Domæne (packages/domain) — forretningsregler

packages/domain/src/absence-codes.ts

Default absence codes for a new Danish tenant (docs/spec/absence-flex.md).

`AbsenceCodeDef` `interface`	Default absence codes for a new Danish tenant (docs/spec/absence-flex.md).

packages/domain/src/access.ts

Access policy (docs/spec/overview.md §3 role matrix, decision D-025) — GRASP Information Expert + Protected Variations: the one place that knows which role (or extra right) may do which action.

`RIGHTS` `const`	Extra rights granted per person on top of roles (spec: "create if right").
`CAPABILITIES` `const`	Actions guarded by the matrix.
`FINANCIAL_AUDIT_ENTITIES` `const`	Audit entities Finance may read ("financial" in the role matrix); Management and Admin read everything.
`AccessPolicy` `class`	What one signed-in person may do, from their roles and extra rights.
`MASKED_ABSENCE` `const`	What a sensitive absence is shown as to people who may not see the reason.

packages/domain/src/budget.ts

Budget vs actual and contribution margin.

`WARNING_BP` `const`	Health thresholds (D-013 / projects spec): warning from 90 %, over above 100 %.
`budgetActual(i: BudgetActualInput)`	Budget vs actual for a project or task: remaining, consumption % (basis points), contribution margin and health.

packages/domain/src/calendar-import.ts

Calendar import rules (docs/spec/integrations.md R6) — GRASP Information Expert for turning calendar events into timesheet suggestions: all-day and "free" events are skipped, events spanning midnight are split per day, and hours are rounded to the nearest 0,25 h (never below 0,25 h for a real meeting).

`CalendarEventInput` `interface`	A calendar event with local wall-clock times in Europe/Copenhagen ("YYYY-MM-DDTHH:mm[:ss…]").
`EventDay` `interface`	One day's part of an event, as a timesheet suggestion.
`roundToQuarter(minutes: number)`	Rounds minutes to centi-hours on the 0,25 h grid; any positive duration becomes at least 0,25 h.
`isImportable(ev: CalendarEventInput)`	True when an event should become a suggestion at all (not all-day, not "free", not cancelled).
`splitEventByDay(ev: CalendarEventInput)`	Splits an event into one suggestion per day it covers (empty when not importable).

packages/domain/src/completion.ts

Fixed-price completion and recognised revenue.

`completionBasisPoints(used: CentiHours, budget: CentiHours, overrideBp?: number)`	Fixed-price percentage of completion in basis points: used ÷ budget hours (centi-hours), capped at 100 %.
`recognisedRevenue(price: Minor, completionBp: number)`	Revenue recognised on a fixed-price contract: price (øre) × completion (basis points), rounded half away from zero.

packages/domain/src/data-import.ts

Data import rules (decision D-024) — GRASP Information Expert for reading customer, project and employee lists from CSV (as saved from Excel): parsing (`;`, `,` or tab, quotes, BOM), Danish/English column names, Danish numbers and dates, and per-row validation.

`ImportKind` `type`	What can be imported.
`ImportColumn` `interface`	One column: the key we use, accepted header names (lower-case), whether it is required, and an example.
`IMPORT_COLUMNS` `const`	Columns per kind.
`parseCsv(text: string)`	Splits CSV text into rows of cells: strips a BOM, detects `;`, tab or `,` from the header, honours quotes.
`parseNumber(v: string | undefined)`	"1.234,50" / "1234.5" / "950" → number, or null when empty; NaN when not a number.
`parseYesNo(v: string | undefined)`	"ja/nej", "yes/no", "x", "1/0", "sand/falsk" → boolean; null when empty; undefined when not understood.
`parseDate(v: string | undefined)`	"2026-12-01", "01-12-2026", "1.12.2026" or "01/12/2026" → "2026-12-01"; null when empty or not a real date.
`MappedImport` `interface`	Result of reading a file: rows keyed by column key (raw strings), the source line numbers, and problems.
`mapImport(kind: ImportKind, table: string[][])`	Maps the header row to column keys (case-insensitive, Danish or English names) and returns keyed rows.
`validateRow(kind: ImportKind, v: Record<string, string>)`	Problems found in one row, as error codes (translated in the UI as `imp.err.<code>`).
`contractTypeOf(v: string)`	"regning" / "fastpris" / "rammeaftale" (or English) → contract type; null when not understood.
`projectStatusOf(v: string)`	"aktiv" / "planlagt" / "afsluttet" / "arkiveret" (or English) → project status; null when not understood.
`rolesOf(…)`	"projektleder, leder" → roles (employee is always added by the importer); null when a name is unknown.
`importTemplate(kind: ImportKind)`	A template CSV (header + one example row, `;`-separated, with BOM so Excel opens it with æøå intact).

packages/domain/src/dates.ts

Local calendar dates as "YYYY-MM-DD" strings in the tenant time zone (Europe/Copenhagen).

`LocalDate` `type`	Local calendar dates as "YYYY-MM-DD" strings in the tenant time zone (Europe/Copenhagen).
`toUtc(d: LocalDate)`	Parses a LocalDate ("YYYY-MM-DD") into a Date at UTC midnight.
`fromUtc(dt: Date)`	Formats a Date as a LocalDate ("YYYY-MM-DD") using its UTC calendar day.
`addDays(d: LocalDate, n: number)`	Adds (or subtracts) whole days to a LocalDate.
`daysBetween(from: LocalDate, to: LocalDate)`	Number of days from `from` to `to` (negative if `to` is earlier).
`isoWeekday(d: LocalDate)`	ISO weekday: Monday = 1 … Sunday = 7.
`startOfIsoWeek(d: LocalDate)`	Monday of the ISO week containing the date.
`isoWeek(d: LocalDate)`	ISO-8601 week number and week-year.
`daysInRange(from: LocalDate, toInclusive: LocalDate)`	All LocalDates from `from` to `toInclusive`, in order.
`todayCopenhagen(now: Date = new Date()`	Today's date in Europe/Copenhagen.

packages/domain/src/expenses.ts

Expenses and mileage (brief §4.7, docs/spec/expenses.md) — GRASP Information Expert for VAT splitting, mileage allowance and the amount re-invoiced to the customer.

`VatSplit` `interface`	Result of splitting a gross (VAT-inclusive) amount.
`vatSplit(gross: Minor, vatBp: number, deductibleBp = 10_000)`	Splits a VAT-inclusive amount into net and VAT, and the VAT into deductible and non-deductible parts.
`expenseCost` `const`	The company's cost of an expense: net amount + VAT it cannot deduct (øre).
`billableAmount(net: Minor, markupBp: number)`	Amount re-invoiced to the customer for a billable expense: net amount plus markup (VAT is added on the invoice).
`MileageRate` `interface`	Mileage rates for one year, øre per km; the high rate applies up to `thresholdKm` per person per year.
`mileageAmount(metres: number, metresBefore: number, rate: MileageRate)`	Mileage allowance for one trip, taking the yearly threshold into account: kilometres before the person reaches the threshold use the high rate, the rest the low rate.
`EXPENSE_STATUSES` `const`	Expense lifecycle — the same as time entries (brief §4.7).
`ExpenseStatus` `type`	One of EXPENSE_STATUSES.

packages/domain/src/flex.ts

Flex.

`weeklyFlex(w: FlexWeekInput)`	Flex for one week in centi-hours: work + absence that counts toward norm − norm hours.
`flexBalance(opening: CentiHours, weeks: FlexWeekInput[])`	Flex balance in centi-hours: opening balance + the flex of every given week.

packages/domain/src/gdpr.ts

GDPR rules (docs/spec/administration.md R5, decision D-026) — GRASP Information Expert for retention and pseudonymisation.

`RETENTION_YEARS` `const`	Years bookkeeping data is retained after the end of its fiscal year.
`retentionCutoff(today: LocalDate, years = RETENTION_YEARS)`	First date that is still inside retention: anything dated before it has passed retention.
`pseudonymFor(userId: string)`	The pseudonym an erased employee gets: stable for the same person (so reports still group correctly) but not reversible.
`ErasureBlocker` `type`	Why an erasure cannot run yet (shown to the Admin, translated as `gdpr.block.<code>`).
`erasureBlockers(p: {…})`	Blockers for erasing a person (spec: must be inactive; not PM on active projects; no one reports to them).

packages/domain/src/holidays-dk.ts

Built-in Danish public holidays (docs/spec/organisation.md).

`easterSunday(year: number)`	Easter Sunday for a year (anonymous Gregorian algorithm).
`danishPublicHolidays(year: number)`	Statutory Danish public holidays for a year, sorted by date (Store bededag only up to 2023).
`defaultCompanyDays(year: number)`	Default extra company days off (decision D-013); tenants can change these per calendar.

packages/domain/src/hours.ts

Hours are handled as integer hundredths of an hour ("centi-hours") to avoid floating-point drift.

`CentiHours` `type`	Hours are handled as integer hundredths of an hour ("centi-hours") to avoid floating-point drift.
`HoursParseError` `class`	Thrown by `parseHours` when the text is not a valid number of hours.
`parseHours(input: string)`	Parses "1,5", "1.5", "1:30", "2" into centi-hours.
`isOnStep(h: CentiHours, step: CentiHours = HOURS_STEP)`	True when the centi-hours are a whole multiple of the step (default 25 = 0,25 h).
`roundToStep(h: CentiHours, step: CentiHours = HOURS_STEP)`	Rounds to the nearest step; ties round up.
`formatHours(h: CentiHours, locale: "da" | "en" = "da")`	Formats centi-hours with two decimals.
`toDecimal` `const`	Centi-hours → decimal hours.
`fromDecimal` `const`	Decimal hours → centi-hours (rounded).

packages/domain/src/invoicing.ts

Invoice suggestion builder, write-down spreading and totals (brief §4.10, docs/spec/invoicing.md).

`isInvoiceable` `const`	Only approved, billable entries not yet on an invoice are invoiceable.
`buildTimeLines(…)`	Builds invoice lines from invoiceable time entries, grouped per task, employee, task + employee, or one summary line.
`spreadAmount(sources: SourceAllocation[], newAmount: Minor)`	Spreads a changed line amount over its sources proportionally to their registered value, using the largest remainder method so the parts always add up exactly to the new amount.
`invoiceTotals(lines: TotalsLine[])`	VAT is calculated per VAT rate on the summed base (not per line) to avoid rounding drift.
`dueDate` `const`	Due date from invoice date and payment terms in days.
`lineAmount(quantity: number, unitPrice: Minor, discountBp = 0)`	Line amount from quantity (centi-hours or centi-units), unit price and discount (basis points).
`AMOUNT_LINE_KINDS` `const`	Invoice line kinds that carry an amount (count in totals/VAT); headings, texts and subtotals do not.
`isAmountLine` `const`	True for line kinds that carry an amount.

packages/domain/src/kpi.ts

Key figures (brief §4.13, docs/spec/reporting.md) — GRASP Information Expert: one definition of each KPI so the dashboards, reports, exports and the Python analytics service (which mirrors these formulas and is tested against the same examples) all agree.

`utilisation` `const`	Utilisation ("udnyttelsesgrad"): billable hours ÷ norm hours.
`billablePct` `const`	Billable share ("faktureringsgrad"): billable ÷ all registered work hours (absence excluded).
`effectiveRate(invoiced: Minor, hours: CentiHours)`	Effective hourly rate: invoiced amount ÷ invoiced hours, in øre per hour (null without hours).
`margin(revenue: Minor, cost: Minor)`	Contribution margin ("dækningsbidrag") and margin % ("DB-grad").

packages/domain/src/money.ts

Money is integer minor units (øre) plus an ISO currency.

`assertMinor(n: number)`	Asserts that a number is a safe integer amount in minor units (øre); throws RangeError otherwise.
`divRound(numerator: number, denominator: number)`	Integer division with rounding half away from zero.
`hoursTimesRate(hours: CentiHours, ratePerHour: Minor)`	hours × hourly rate → amount in minor units.
`percentOf(amount: Minor, basisPoints: number)`	amount × percentage in basis points (2500 = 25,00 %).
`formatMinor(amount: Minor, locale: "da" | "en" = "da")`	Formats øre as "1.234,56" (da) or "1,234.56" (en), without currency symbol.
`kr` `const`	kr → øre for literals in tests and seed data: kr(925) = 92500.

packages/domain/src/norm.ts

Norm (expected) hours.

`FULL_TIME_37` const	Default Danish full time: Mon–Thu 7,5 h, Fri 7,0 h = 37 h.
`patternHours(date: LocalDate, a: NormAssignment)`	Norm hours (centi-hours) a pattern gives for a date, before holidays; handles 1- and 2-week cycles.
`dailyNorm(date: LocalDate, ctx: NormContext)`	Norm hours for one day after the employment period and public holidays.
`normForRange(from: LocalDate, days: number, ctx: NormContext)`	Sum of daily norm hours (centi-hours) for `days` consecutive days starting at `from`.
`weeklyNorm` const	Norm hours (centi-hours) for the ISO week starting on `monday`, after holidays and employment dates.

packages/domain/src/payroll.ts

Payroll export rules (docs/spec/payroll.md, decision D-023) — GRASP Information Expert for turning approved absence, mileage and expenses into payroll lines.

`SalaryUnit` type	Unit of a salary code ("lønart").
`absenceDayFraction(hours: CentiHours, normThatDay: CentiHours)`	Absence as days (R2): hours ÷ the employee's norm that day, in hundredths of a day.
`PayrollLine` interface	One payroll line: an employee's total on one salary code for one month (or a late delta for an earlier month).
`lineKey` const	Key identifying employee × salary code × month.
`lateDeltas(current: PayrollLine[], exported: PayrollLine[])`	Late items (R7): for months that are already locked, the difference between what the data says now and what was exported in locked runs.
`PAYROLL_CSV_HEADER` const	Columns of the generic CSV (R9).
`PayrollCsvRow` interface	One CSV row's input.
`payrollCsv(rows: PayrollCsvRow[])`	Generic payroll CSV (R9): UTF-8 with BOM, `;` separator, Danish decimal comma, ISO dates, CRLF line ends.

packages/domain/src/periods.ts

Accounting periods (brief §4.11, docs/spec/accounting-periods.md): monthly per legal entity, open → closed.

`monthOf` const	"2026-11-08" → "2026-11-01"
`isLocked(date: LocalDate, closedMonths: ReadonlySet<LocalDate>)`	True when the date falls in a closed accounting month (`closedMonths` holds first-of-month dates).
`lastDayOfMonth(month: LocalDate)`	Last day of the month as a LocalDate.
`recognitionDelta(recognisedToDate: Minor, previouslyRecognised: Minor)`	Month-end figures for one fixed-price contract: revenue recognised to date = price × completion; revenue to recognise this month = recognised to date − previously recognised (may be negative after a budget change).
`deferredRevenue(invoiced: Minor, usedHours: number, poolHours: number)`	Deferred revenue on a prepaid retainer: invoiced − earned (never negative; an overrun is WIP, not negative deferral).

packages/domain/src/planning.ts

Resource planning (brief §4.8, docs/spec/resource-planning.md) — GRASP Information Expert for capacity and utilisation bands.

`UtilisationBand` `type`	Heatmap band (brief §4.8): under < 70 %, ok 70–100 % (both inclusive), over > 100 %; none = no capacity.
`capacity(normHours: CentiHours, absenceHours: CentiHours)`	Capacity for a week: norm − absence, never below zero.
`utilisationBp(booked: CentiHours, cap: CentiHours)`	Booked ÷ capacity in basis points (null when there is no capacity).
`utilisationBand(booked: CentiHours, cap: CentiHours)`	Heatmap colour band for a week.
`freeHours` `const`	Free hours in a week (capacity − booked, never negative).

packages/domain/src/rates.ts

Hourly rate resolution (brief §4.9, docs/spec/rates.md).

`matchPriceList(…)`	Most specific matching line of one price list, or undefined.
`resolveRate(c: RateContext)`	Resolves the hourly sales rate (øre) for one time entry using the brief's 5-step order and returns which step won.

packages/domain/src/roles.ts

Roles are additive (docs/spec/overview.md §3).

`ROLES` `const`	Roles are additive (docs/spec/overview.md §3).
`hasRole` `const`	True when the user holds at least one of the wanted roles.

packages/domain/src/timesheet.ts

Time-entry validation and weekly summary.

`isEditableStatus` `const`	Entries the employee may still edit: draft and rejected (rejected returns to draft on edit).
`validateEntry(c: EntryCheck)`	Checks one time-entry change against the timesheet rules and returns every error found (empty = valid).
`summariseWeek(…)`	Week summary for one employee: per-day norm/work/billable/absence and week totals, flex and billable % (basis points).
`dayAnomalies(…)`	Anomaly flags for the approver inbox (docs/spec/approval.md).

packages/domain/src/vacation.ts

Danish vacation (samtidighedsferie).

`holidayYearOf(d: LocalDate)`	The holiday year (start year) that a date belongs to: 2026-11-08 → 2026 (1 Sep 2026 – 31 Aug 2027).
`accruedDays(…)`	Days earned in holiday year `hy` up to and including `asOf`, for an employee employed from `start` (to optional `end`).
`vacationBalance(opening: CentiDays, accrued: CentiDays, used: CentiDays)`	Vacation balance = opening + accrued − used (all in centi-days).

Database (packages/db) — skema, RLS og arbejdsgange

packages/db/src/audit.ts

Audit trail writer (brief §4.15: every change to financial data is logged — who, when, before → after, reason).

`writeAudit(tx: Tx, e: AuditEntry)`	Writes an audit record inside the caller's transaction, so the change and its audit trail commit together.

packages/db/src/client.ts

Database connection and tenant scoping (decision D-012).

`createDb(url: string, opts: { max?: number } = {})`	Creates a Drizzle client over a postgres.js pool; `close()` ends the pool.
`withTenant(db: Db, tenantId: string, fn: (tx: Tx)`	Runs `fn` in a transaction scoped to one tenant.

packages/db/src/invoicing.ts

Invoicing service (brief §4.10, docs/spec/invoicing.md, decisions D-014/D-015).

`InvoiceError` `class`	Business error from the invoicing service; `code` is shown to users via i18n, `status` is the HTTP status.
`closedMonths(tx: Tx)`	Closed accounting months for the tenant (single legal entity in MVP; per entity in M4).
`invoiceableEntries(…)`	Approved, billable, unreserved T&M entries for the given projects up to a date.
`createDraftFromSuggestion(tx: Tx, p: DraftParams)`	Creates a draft invoice from everything invoiceable on the projects (one customer).
`loadInvoice(tx: Tx, invoiceId: string)`	Loads an invoice header by id or throws InvoiceError("not_found", 404).
`recalcInvoice(tx: Tx, invoiceId: string)`	Recomputes subtotal, VAT (per rate) and total from the invoice's amount lines and stores them.
`updateLine(tx: Tx, tenantId: string, userId: string, lineId: string, u: LineUpda…)`	Edits a draft invoice line.
`addFreeLine(…)`	Adds a manual line (product, text, heading or subtotal) at the end of a draft and recalculates totals.
`deleteLine(tx: Tx, lineId: string)`	Deletes a draft line and releases its reserved time entries / milestones so they become invoiceable again.
`deleteDraft(tx: Tx, tenantId: string, userId: string, invoiceId: string)`	Deletes an unbooked invoice, releasing all reserved entries and milestones; audited.
`bookInvoice(…)`	Books an invoice (D-015): locks the legal entity's counter row, takes the next number, marks sources invoiced, writes audit + outbox — all in the caller's transaction.
`createCreditNote(…)`	Creates a draft credit note for a booked invoice: all lines, or the chosen ones, negated.
`isDateOpen(tx: Tx, date: LocalDate)`	True when a date may still be changed (its accounting month is open).

packages/db/src/migrate.ts

CLI entry point: applies all SQL migrations in packages/db/drizzle as the owner role (`pnpm db:migrate`).

packages/db/src/periods.ts

Accounting periods and month-end figures (brief §4.11, docs/spec/accounting-periods.md).

`PeriodError` `class`	Business error from the accounting-period service; `code` for i18n, `status` for HTTP.
`ensurePeriods(…)`	Makes sure a period row exists for every month from `fromMonth` to `toMonth`.
`monthEndReport(tx: Tx, month: LocalDate)`	Month-end figures for a month (first day as LocalDate): WIP on T&M (approved, not on a booked invoice), fixed-price recognition (to date, this month, invoiced to date incl.
`closeChecklist(tx: Tx, month: LocalDate)`	Pre-close checklist: unapproved timesheets and unbooked invoices dated in the month.
`closeMonth(…)`	Closes an accounting month if all earlier months are closed.
`reopenMonth(…)`	Reopens the most recently closed month only (audited).

packages/db/src/rates.ts

Loads rate-resolution inputs from the database and snapshots sales/cost rates on time entries (brief §4.9, decision D-014).

`snapshotRates(tx: Tx, entryIds: string[])`	Resolves and stores the sales rate (billable entries only) and cost rate (øre/h) on the given time entries.

packages/db/src/schema.ts

Database schema.

`priceGroups` `const`	Rate role of an employee (e.g.
`paymentPlanItems` `const`	Fixed-price payment plan / retainer fee schedule.
`numberSeries` `const`	Gap-free invoice numbering per legal entity (brief §4.10).
`invoiceLineSources` `const`	Which time entries an invoice line covers, with registered and invoiced value (write-down analytics).
`revenueRecognitions` `const`	Snapshot taken when a month is closed: fixed-price revenue recognised to date and the month's delta.
`integrationConnections` `const`	Configured accounting/payroll connections.
`outbox` `const`	Transactional outbox: events written in the same transaction as the change, delivered by a worker.
`expenseTypes` `const`	Expense categories; `vatDeductibleBp` is the share of the VAT the company may deduct (e.g.
`mileageRates` `const`	Mileage allowance per year: high rate (øre/km) up to the threshold, low rate above.
`files` `const`	Uploaded files (receipts).
`expenses` `const`	An out-of-pocket expense or a mileage claim.
`allocations` `const`	Planned hours: one employee on one project in one ISO week (decision OQ-28 default: weekly granularity).
`userCompetencies` `const`	Competencies as simple tags per employee (e.g.
`reminderLog` `const`	E-mail reminders already sent, so the weekly job never sends the same reminder twice.
`graphTokens` `const`	Delegated Microsoft Graph access token per user (read-only scope Calendars.Read), AES-GCM encrypted with a key derived from SESSION_SECRET.
`calendarImports` `const`	Calendar events (per day) the user has imported into the timesheet or chosen to ignore — shown as handled.
`salaryCodes` `const`	Salary codes ("lønarter") the payroll system knows.
`payPeriods` `const`	Monthly pay periods: open → exported → locked.
`payrollRuns` `const`	One generated export file for a pay period.
`payrollLines` `const`	Lines of a run; `month` is the month the items are dated in (earlier than the run's month for late lines).
`userRights` `const`	Extra rights on top of roles (D-025), e.g.
`apiTokens` `const`	Personal read-only tokens for the analytics service (spec integrations R10, decision D-028), e.g.

packages/db/src/seed-access.ts

Fictional seed for extra rights (D-025): Anne Lund (AL) and Peter Dahl (PD) may create projects; the other project managers may not.

`seedAccess(tx: Tx, tenantId: string)`	Grants the demo rights once.

packages/db/src/seed-expenses.ts

M3 part of the fictional seed: expense types, mileage rates and a handful of expenses/mileage claims in the open months (so they show up in approval and in the invoicing basis).

`seedExpenses(tx: Tx, tenantId: string, today: LocalDate)`	Seeds expense types, mileage rates and example expenses for the demo tenant (fictional data only).

packages/db/src/seed-money.ts

M2 part of the fictional seed: rate groups, task types, price lists, payment plans, booked invoice history for the first months after go-live, and closed accounting periods.

`seedMoney(…)`	Seeds the money part of the fictional demo company: price groups, task types, price lists, rate snapshots, payment plans, booked invoice history and closed months.

packages/db/src/seed-payroll.ts

M4 part of the fictional seed: salary codes ("lønarter") with illustrative provider codes, the mapping from absence codes, and payroll numbers for every employee except Frederik (FH) — so the export shows the "missing payroll number" warning.

`seedPayroll(tx: Tx, tenantId: string)`	Seeds payroll setup for one tenant (no-op when already present).

packages/db/src/seed-planning.ts

M3 part of the fictional seed: competencies per employee and allocations for the coming weeks, spread over each employee's customer projects at roughly 55–115 % of a 37-hour week so the heatmap shows all bands.

`seedPlanning(tx: Tx, tenantId: string, today: LocalDate)`	Seeds competencies and 8 weeks of allocations from the current ISO week (fictional data only).

packages/db/src/seed.ts

Fictional seed data: "ARCnordic Demo ApS" (brief §1 and §9).

API-services (apps/api/src/services) — arbejdsgange

apps/api/src/services/api-tokens.ts

Personal API tokens for the analytics service (spec integrations R10, decision D-028) — GRASP Pure Fabrication.

`listTokens(tx: Tx, auth: AuthContext)`	The caller's tokens, newest first (never the token itself).
`createToken(tx: Tx, auth: AuthContext, name: string)`	Creates a read-only analytics token for the caller; returns the raw token once.
`revokeToken(tx: Tx, auth: AuthContext, id: string)`	Revokes one of the caller's tokens (Power BI stops refreshing at once).

apps/api/src/services/approval.ts

Approval service (docs/spec/approval.md, decision D-013: the line manager approves the week) — GRASP Pure Fabrication for the approval workflows.

`approvalInbox(tx: Tx, auth: AuthContext, weekStart: LocalDate)`	The approver's inbox for one ISO week: every person the caller approves with status, norm, registered hours, billable %, flex this week, anomaly flags (days far over/under norm) and hours per project.
`approvePeriod(tx: Tx, auth: AuthContext, periodId: string)`	Approves a submitted week: submitted entries → approved, rate snapshot taken (5-step resolution), period → approved, event logged and audited.
`rejectPeriod(tx: Tx, auth: AuthContext, periodId: string, comment: string)`	Rejects a submitted week with a required comment shown to the employee; entries → rejected (editable again).
`reopenApprovedPeriod(tx: Tx, auth: AuthContext, periodId: string, reason: string)`	The approver reopens an approved week (e.g.

apps/api/src/services/audit-log.ts

Audit log reader (docs/spec/administration.md R1–R2, decision D-025) — GRASP Pure Fabrication.

`AuditFilter` `interface`	Filters for the audit screen; `before` is the cursor (the `at` of the last row already shown).
`auditEntities(tx: Tx, auth: AuthContext)`	Entity types the caller may filter on (only those present in the log).
`readAuditLog(tx: Tx, auth: AuthContext, f: AuditFilter)`	Newest-first page of audit rows the caller may read, with the actor's name.

apps/api/src/services/calendar.ts

Loads an employee's norm context and computes flex and vacation balances from the database.

`loadEmployee(tx: Tx, userId: string)`	Loads the user row with employment, rates and opening balances; undefined if not found (RLS-scoped).
`normContext(tx: Tx, emp: Employee)`	Builds the domain NormContext for an employee: norm assignments with valid-from, holiday calendar, employment dates.
`summaryEntries(…)`	Entries of one employee in a date range, in the shape the domain summary needs.
`flexBalance(…)`	Flex balance (docs/spec/absence-flex.md): opening balance at go-live + Σ weekly flex for completed weeks (setting "computed up to last Sunday", D-013) — the current week is shown separately as "this week".
`vacationBalance(tx: Tx, emp: Employee, today: LocalDate)`	Vacation (samtidighedsferie, D-013): opening + accrued − used in the current holiday year, in centi-days.

apps/api/src/services/customers.ts

Customers (docs/spec/crm.md, light CRM) — GRASP Information Expert for customer records: listing with project counts and creating customers with the next customer number.

`listCustomers(tx: Tx)`	All customers with their number of projects, sorted by name.
`NewCustomerInput` `interface`	Input for a new customer (CVR 8 digits, EAN 13 digits for public-sector e-invoicing).
`createCustomer(tx: Tx, auth: AuthContext, input: NewCustomerInput)`	Creates a customer with the next number in the customer series (1001, 1002 …) and audits it.

apps/api/src/services/dashboard.ts

Role-based home dashboard and reminders (brief §4.13, §4.6 reminders) — GRASP Pure Fabrication: gathers, per role, the few things the person should act on ("to-dos") and the key figures they steer by.

`Todo` `interface`	One item on the person's to-do list; `count`/`ref` fill the translated text, `href` is where to act.
`WeekKpi` `interface`	Week KPI row (decimals for the UI, basis points for percentages).
`dashboard(tx: Tx, auth: AuthContext, today: LocalDate)`	The dashboard for the signed-in person: their to-dos (by role) and, for line managers, Finance, Management and Admin, key figures — weekly utilisation/billable % for the people they may see (12 weeks), and invoiced revenue with effective hourly rate for this and last month (Finance/Management/Admin only).

apps/api/src/services/data-import.ts

Data import (decision D-024) — GRASP Pure Fabrication: turns a CSV of customers, projects or employees into a plan (create / update per row, with errors), shows it as a dry run, and applies it in one transaction — all rows or none.

`PlannedRow` `interface`	One planned row: what will happen and why not, if it cannot.
`previewImport(tx: Tx, kind: ImportKind, csv: string)`	Dry run: what the import would do, row by row, without writing anything.
`commitImport(tx: Tx, auth: AuthContext, kind: ImportKind, csv: string)`	Applies the import (refused with 422 if any row has errors).

apps/api/src/services/errors.ts

Application error used by services and controllers alike (GRASP: Low Coupling — services do not depend on the HTTP/auth layer).

`AppError` `class`	Application error used by services and controllers alike (GRASP: Low Coupling — services do not depend on the HTTP/auth layer).

apps/api/src/services/expenses.ts

Expenses and mileage (brief §4.7, docs/spec/expenses.md) — GRASP Pure Fabrication for the expense workflows: register (draft), submit, approve/reject, list.

`expenseSettings(tx: Tx)`	Expense types and mileage rates for the registration forms.
`myExpenses(tx: Tx, auth: AuthContext, month: LocalDate)`	The signed-in employee's expenses in a month (`month` = first day, "YYYY-MM-01"), newest first.
`ExpenseInput` `interface`	Input for an expense (`kind: "expense"`, gross amount in kr incl.
`saveExpense(tx: Tx, auth: AuthContext, input: ExpenseInput, id?: string)`	Creates or updates (when `id` is given and still draft/rejected) an expense for the signed-in employee and computes its amounts: VAT split by expense type for expenses; allowance from the yearly rate table for mileage.
`deleteExpense(tx: Tx, auth: AuthContext, id: string)`	Deletes one of the employee's own draft/rejected expenses.
`submitExpenses(tx: Tx, auth: AuthContext, ids?: string[])`	Submits the employee's draft/rejected expenses (all of them, or the given ids) for approval.
`expenseApprovalQueue(tx: Tx, auth: AuthContext)`	Submitted expenses waiting for the caller: their reports' (line manager) or everyone's (Finance, D-025).
`approveExpense(tx: Tx, auth: AuthContext, id: string)`	Approves a submitted expense (audited).
`rejectExpense(tx: Tx, auth: AuthContext, id: string, comment: string)`	Rejects a submitted expense with a required comment shown to the employee (audited).
`canSeeFile(tx: Tx, auth: AuthContext, fileId: string)`	True if the caller may open the receipt file: its owner, the approver of the expense, or Finance/Admin.

apps/api/src/services/files.ts

File storage for receipts (brief §4.7) — GRASP Pure Fabrication + Protected Variations: callers only see `storeFile` / `readFile`; today bytes are kept on the local disk (`data/uploads/<tenant>/<file id>`, D-017 runs everything locally), later this module can switch to Azure Blob/SharePoint without touching callers.

`MAX_FILE_BYTES` `const`	Maximum receipt size in bytes (10 MB).
`sniffContentType(bytes: Buffer)`	Detects the real content type from the first bytes; null when the type is not allowed.
`storeFile(tx: Tx, auth: AuthContext, filename: string, bytes: Buffer)`	Validates and stores an uploaded file for the signed-in user; returns the file row (id used by expenses).
`readStoredFile(tx: Tx, auth: AuthContext, fileId: string)`	Reads a stored file if the caller may see it: its owner, or anyone who may see an expense that references it (checked by the caller via `canSeeFile`).

apps/api/src/services/gdpr.ts

GDPR workflows (docs/spec/administration.md, decision D-026) — GRASP Pure Fabrication: data-subject export, erasure (pseudonymisation) of former employees, and the nightly retention sweep.

`exportPersonalData(tx: Tx, auth: AuthContext, userId: string)`	Everything ARCnordic Tid holds about one person, as one JSON document (spec: data-subject export; AC5 — nothing about others).
`erasurePreview(tx: Tx, userId: string)`	What an erasure would do and whether it is blocked (shown to the Admin before confirming).
`erasePerson(tx: Tx, auth: AuthContext, userId: string, reason: string)`	Erases a former employee (spec R5, AC4): refuses while blocked, then pseudonymises the profile, clears comments on kept entries, and deletes non-bookkeeping personal data.
`retentionSweep(tx: Tx, today: LocalDate)`	Nightly retention sweep for one tenant: deletes short-lived personal data (expired sessions and Graph tokens, reminder and calendar-import logs older than 13 months) and reports bookkeeping records past the 5-year retention, which Finance purges deliberately (D-026) rather than a job deleting accounting data on its own.
`pastRetention(tx: Tx, today: LocalDate)`	Bookkeeping records dated before the retention cutoff, and former employees not yet erased.
`runRetentionSweep(db: Db, today: LocalDate = todayCopenhagen()`	Runs the retention sweep for every tenant (background job; each tenant in its own RLS-scoped transaction).

apps/api/src/services/invoice-document.ts

Invoice document (HTML → PDF, decision D-016).

`invoiceHtml(tx: Tx, invoiceId: string)`	Renders the invoice/credit note as a self-contained Danish HTML document (KLADDE watermark until booked, optional time-specification appendix).
`htmlToPdf(html: string)`	Renders HTML to an A4 PDF with headless Chromium (one shared browser per process).

apps/api/src/services/invoices.ts

Invoice access and read models for the API (brief §4.10, docs/spec/invoicing.md) — GRASP Pure Fabrication.

`assertCanInvoice(tx: Tx, auth: AuthContext, projectIds: string[])`	Throws 403 unless the caller may prepare invoices for all given projects (Finance, or PM of each project).
`assertCanSeeInvoice(tx: Tx, auth: AuthContext, invoiceId: string)`	Loads an invoice the caller may see, or throws 404 (not 403, so existence is not revealed).
`assertCanSeeLine(tx: Tx, auth: AuthContext, lineId: string)`	Loads the invoice that owns a line and checks the caller may see it; returns the invoice id.
`invoicingBasis(tx: Tx, auth: AuthContext, upTo: LocalDate)`	Invoicing basis ("faktureringsgrundlag"): per customer project, everything invoiceable up to `upTo` — approved, billable, unreserved T&M hours (value at the approval rate snapshot, øre), due or ready payment-plan items, and approved billable expenses (net + markup).
`activeCustomerProjects(tx: Tx)`	Active customer projects, for one-click bulk drafting.
`listInvoices(…)`	Invoice list filtered by status ("open" = not booked), newest number first; PMs see their projects only.
`invoiceDetail(tx: Tx, auth: AuthContext, invoiceId: string)`	Full invoice for the editor: header, customer, lines with how many time entries each covers, VAT totals, credit notes.
`InvoiceHeaderUpdate` `interface`	Editable invoice header fields.
`updateInvoiceHeader(…)`	Updates header fields of an unbooked invoice (audited).
`markReady(tx: Tx, auth: AuthContext, invoiceId: string)`	Marks a draft "ready for booking" (lets a PM hand over to Finance; decision D-013 / observed separation of roles).

apps/api/src/services/outbox.ts

Outbox delivery (transactional outbox, docs/spec/integrations.md).

`processOutbox(…)`	Claims up to 20 pending outbox events across tenants and delivers them (e.g.

apps/api/src/services/outlook.ts

Outlook calendar import (docs/spec/integrations.md R6) — GRASP Pure Fabrication + Indirection: shows a week's meetings next to the timesheet and turns the ones the user picks into draft hours on a chosen task.

`CalendarProvider` `interface`	Where the calendar comes from.
`CalendarStatus` `type`	disabled = not configured · not_connected = user must sign in with Microsoft · connected = events listed
`encryptToken(plain: string, secret: string)`	Encrypts a token as "iv.tag.ciphertext" (base64url, AES-256-GCM).
`decryptToken(value: string, secret: string)`	Reverses encryptToken; throws if the value was tampered with or the secret changed.
`storeGraphToken(…)`	Stores (or replaces) the user's delegated Graph token after Entra sign-in.
`graphCalendarProvider(enabled: boolean, secret: string)`	The production provider: Microsoft Graph with the stored token (disabled unless configured).
`CalendarItem` `interface`	One meeting-day as shown in the UI (hours as decimals), with what the user already did with it.
`weekCalendar(…)`	The week's meetings for the signed-in user, split per day and rounded to 0,25 h, marked as new/imported/ignored.
`ImportItem` `interface`	A meeting-day the user wants on the timesheet.
`importCalendarItems(tx: Tx, auth: AuthContext, items: ImportItem[])`	Adds each picked meeting as draft hours on the chosen task (added to what the cell already holds; the meeting subject becomes the comment when the cell has none) and marks it imported.
`ignoreCalendarItem(tx: Tx, auth: AuthContext, eventId: string, date: LocalDate)`	Marks a meeting-day as "not time to register" so it no longer shows as new.

apps/api/src/services/payroll.ts

Payroll export (docs/spec/payroll.md, decision D-023) — GRASP Pure Fabrication: gathers approved absence, mileage and personal expenses per employee × salary code × month, previews them with warnings, writes the generic CSV and locks months.

`payPeriods(tx: Tx, auth: AuthContext)`	Creates monthly pay periods from the go-live month up to the current month (idempotent) and lists them.
`PreviewLine` `interface`	One preview line for the UI (quantities as decimals, amounts in øre).
`payrollPreview(tx: Tx, auth: AuthContext, month: LocalDate)`	Preview of a month's export: one row per employee with their lines and warnings — missing payroll number (row not exported, AC2) and items still awaiting approval (they become late lines once approved).
`exportPayroll(tx: Tx, auth: AuthContext, month: LocalDate)`	Generates (or regenerates) the export for a month: employees without a payroll number are left out, the file replaces any earlier unlocked run (AC4), and the period becomes "exported".
`lockPayPeriod(tx: Tx, auth: AuthContext, month: LocalDate)`	Locks an exported month: its lines are frozen; later changes to it become late lines (R8).
`payrollFile(tx: Tx, month: LocalDate)`	The CSV file of a month's run (404 when not exported yet).

apps/api/src/services/people.ts

Who may see or approve whom (docs/spec/overview.md §3, decision D-013).

`reportIds(tx: Tx, managerId: string)`	Direct and indirect reports of a manager.
`visibleUserIds(tx: Tx, auth: AuthContext)`	Users whose time the caller may read: self, reports (line manager), everyone (admin/finance/management).
`relationTo(…)`	How the caller relates to an employee, for privacy rules (AccessPolicy.seesAbsenceReason): the employee themself, their line manager (direct or indirect), or anyone else.
`canViewUser(tx: Tx, auth: AuthContext, userId: string)`	True if the caller may read the user's time data (self, a report of a line manager, or Admin/Finance/Management).
`canApproveUser(tx: Tx, auth: AuthContext, userId: string)`	Approver rule (D-013, D-025): the direct line manager approves.
`approveeIds(tx: Tx, auth: AuthContext)`	Users the caller approves (for the inbox).

apps/api/src/services/planning.ts

Resource planner (brief §4.8, docs/spec/resource-planning.md) — GRASP Pure Fabrication: builds the capacity heatmap (capacity vs booked per employee per ISO week) and changes allocations.

`PlannerWeek` `interface`	One week cell in the heatmap (hours as decimals for the UI).
`plannerView(…)`	The heatmap for `weeks` ISO weeks from `fromMonday`: per employee their weeks (capacity, booked, free, band), allocation rows per project, competencies; optionally filtered by competency (case-insensitive substring).
`setAllocation(…)`	Sets planned hours (decimal) for an employee on a project in one ISO week; 0 removes the allocation.
`setCompetencies(tx: Tx, auth: AuthContext, userId: string, names: string[])`	Replaces an employee's competency tags (Admin or their line manager).
`allCompetencies(tx: Tx)`	Distinct competency names in the company (for the filter suggestions).

apps/api/src/services/project-approval.ts

Project manager approval step (docs/spec/approval.md, mode pm_then_line_manager, decision D-022) — GRASP Pure Fabrication.

`ApprovalMode` `type`	How weeks are approved in this tenant.
`approvalMode(tx: Tx, tenantId: string)`	The tenant's approval mode (defaults to line manager only).
`pendingPmSteps(tx: Tx, userId: string, weekStart: LocalDate)`	Number of open PM steps (distinct projects) in one employee's week — shown to the line manager.
`pmInbox(tx: Tx, auth: AuthContext)`	The project manager's inbox: per employee × week × project, the submitted hours waiting for them, with the individual entries (date, task, hours, comment).
`PmDecision` `interface`	One PM decision: an employee's week on one project.
`pmDecide(tx: Tx, auth: AuthContext, d: PmDecision)`	Records the project manager's decision.

apps/api/src/services/project-setup.ts

Project setup workflows (docs/spec/projects.md) — GRASP Creator + Pure Fabrication: creating projects (with their first contract and the PM as member), adding tasks to the WBS and editing tasks.

`WBS_MAX_LEVEL` `const`	Maximum WBS depth (brief §4.3).
`NewProjectInput` `interface`	Input for a new project; money in kroner (converted to øre here).
`createProject(tx: Tx, auth: AuthContext, b: NewProjectInput)`	Creates an active project with one contract of the chosen type and the PM as member (audited).
`NewTaskInput` `interface`	Input for a new task; budget amount in kroner.
`addTask(tx: Tx, auth: AuthContext, projectId: string, b: NewTaskInput)`	Adds a task to the project's WBS (max 5 levels).
`TaskUpdate` `interface`	Editable task fields; budget amount in kroner.
`updateTask(tx: Tx, auth: AuthContext, taskId: string, patch: TaskUpdate)`	Updates a task (name, budget, open/closed, billable) — only the project's PM, Finance or Admin (audited).

apps/api/src/services/projects.ts

Project scope and budget-vs-actual aggregation (docs/spec/projects.md, reporting.md).

`visibleProjectIds(tx: Tx, auth: AuthContext)`	Projects the caller may see: all (admin/finance/management), otherwise managed + member projects.
`canEditProject(tx: Tx, auth: AuthContext, projectId: string)`	True if the caller may change the project: Admin/Finance, or its project manager.
`taskActuals(tx: Tx, projectIds: string[])`	Actuals per task.
`projectSummaries(tx: Tx, projectIds: string[] | "all")`	Project list rows with budget vs actual (hours, value, cost, margin, health) for the given ids or all projects.
`projectDetail(tx: Tx, projectId: string)`	Task tree with budget vs actual rolled up from leaves, plus contract completion for fixed price.

apps/api/src/services/reminders.ts

Weekly e-mail reminders (brief §4.6 "reminders") — GRASP Pure Fabrication + Indirection: a job that, on Monday mornings (Europe/Copenhagen), reminds employees who have not submitted last week and approvers who have timesheets waiting.

`Mailer` `interface`	Minimal mail-sending interface (nodemailer in production, a fake in tests).
`mailerFromEnv(env: Record<string, string | undefined>)`	Builds an SMTP mailer from env (SMTP_HOST/SMTP_PORT/MAIL_FROM), or null when not configured — the default.
`isReminderWindow(now: Date = new Date())`	True from Monday 09:00 Copenhagen time until the end of Monday — the window the weekly job may run in.
`sendTenantReminders(…)`	Sends this week's reminders for one tenant; returns how many e-mails were sent.
`runWeeklyReminders(…)`	Runs the weekly reminders for every tenant if we are in the Monday window and a mailer is configured.

apps/api/src/services/reports.ts

Reports and balances (docs/spec/reporting.md) — GRASP Pure Fabrication: report queries and CSV export.

`toCsv(header: string[], rows: (string | number | null)`	Builds a CSV document for Danish Excel.
`myBalances(tx: Tx, auth: AuthContext)`	The signed-in employee's flex balance (hours, completed weeks) and vacation balance (days) for the holiday year.
`budgetActualRows(tx: Tx, auth: AuthContext)`	Budget vs actual for every project the caller may see (same figures as the project list).
`budgetActualCsv(rows: Awaited<ReturnType<typeof budgetActualRows>>)`	Budget vs actual as a Danish CSV document.
`TimeEntryFilter` `interface`	Filters for the time-entry report.
`timeEntryRows(tx: Tx, auth: AuthContext, f: TimeEntryFilter)`	Time entries in a date range for the people the caller may see (self; reports for a line manager; everyone for Finance/Admin/Management).
`timeEntryCsv(rows: Awaited<ReturnType<typeof timeEntryRows>>)`	Time entries as a Danish CSV document.

apps/api/src/services/timesheet.ts

Timesheet service (docs/spec/time-tracking.md) — GRASP Pure Fabrication: the database workflows behind the weekly timesheet.

`LOCKED_PERIOD` `const`	Timesheet period statuses in which the employee can no longer edit entries.
`tenantGoLive(tx: Tx, tenantId: string)`	The tenant's go-live date: balances are opening balance at this date + registrations after it.
`getPeriod(tx: Tx, userId: string, weekStart: LocalDate)`	The employee's timesheet period for the ISO week starting `weekStart` (Monday), or undefined.
`ensurePeriod(tx: Tx, auth: AuthContext, userId: string, weekStart: LocalDate)`	Returns the period for the week, creating an "open" one if none exists (GRASP Creator: the week owns its period).
`taskRows(tx: Tx, taskIds: string[])`	Task + project information for the given task ids (used for row labels and entry validation).
`weekView(tx: Tx, auth: AuthContext, userId: string, weekStart: LocalDate)`	Builds everything the week screen shows for one employee and week: rows (task/absence) with cells per day, norm per day, holidays, closed days, week summary (hours, billable %, flex this week), flex balance and the event log.
`CellInput` `interface`	One cell of the timesheet: a task OR an absence code on a date, with decimal hours (0 clears the cell).
`saveCell(tx: Tx, auth: AuthContext, b: CellInput)`	Saves one timesheet cell for the signed-in employee: checks membership and leaf task, runs the shared domain validation (24 h/day, 0,25 step, future work, closed project/task, required comment, locked period), then creates, updates or clears the entry.
`submitWeek(tx: Tx, auth: AuthContext, weekStart: LocalDate, comment: string | nu…)`	Submits the employee's week for approval: draft/rejected entries → submitted, period → submitted, event logged.
`withdrawWeek(tx: Tx, auth: AuthContext, weekStart: LocalDate)`	Lets the employee take back a submitted week before anyone has approved it (audited).
`assignableTasks(tx: Tx, auth: AuthContext, search?: string)`	Tasks the employee can register on: open leaf tasks on active projects they are a member of; optional text search.
`addFavourite(tx: Tx, auth: AuthContext, taskId: string)`	Marks a task as one of the employee's favourites (idempotent).
`removeFavourite(tx: Tx, auth: AuthContext, taskId: string)`	Removes a task from the employee's favourites.
`previousWeekRows(tx: Tx, auth: AuthContext, weekStart: LocalDate)`	The still-usable task rows the employee had the week before `weekStart` ("copy last week's rows" — rows only, no hours).

API-ruter og login (apps/api/src/routes, auth)

apps/api/src/app.ts

Fastify application factory — wires security plugins (helmet, CORS, cookies, rate limit), OpenAPI docs, health checks, the auth plugin and all /api/v1 route modules, plus the global error handler that turns AppError, invoice/period errors and Postgres check violations into `{ error: code }` responses.

`AppDeps` `interface`	Replaceable collaborators (tests pass fakes); defaults are the real, config-driven implementations.
`buildApp(config: Config, db: Db, deps: AppDeps = {})`	Builds the configured Fastify app (no listening) — injected config and database make it testable with `app.inject`.

apps/api/src/auth/plugin.ts

Authentication + CSRF.

`AuthContext` `interface`	The signed-in person for one request.
`policyOf` `const`	The access policy for a request's person (the role matrix lives in packages/domain/src/access.ts).
`authPlugin` `const`	Fastify plugin: resolves the session cookie to `request.auth`, adds `request.withTenant()`, and enforces login and the CSRF header on every route not marked `config: { public: true }`.
`requireRole(req: FastifyRequest, ...roles: Role[])`	Returns the auth context if the caller has at least one of the roles, otherwise throws 401/403.
`requireCapability(req: FastifyRequest, capability: Capability)`	Throws 401/403 unless the caller's access policy allows the capability; returns the auth context.

apps/api/src/auth/routes.ts

Authentication routes — GRASP Controller: dev login (email + password, never in production), logout, and Microsoft Entra ID sign-in via OIDC with PKCE (enabled only when configured).

`authRoutes` `const`	Registers /auth/providers, /auth/dev-login (dev only), /auth/logout and the Entra ID login/callback routes.

apps/api/src/auth/session.ts

Session store — opaque random tokens in an httpOnly cookie; only their SHA-256 hash is stored in the database, together with a per-session CSRF token.

`sha256` `const`	Hex SHA-256 of a string (used so raw session tokens are never stored).
`createSession(…)`	Creates a 12-hour session for the user in their tenant; returns the raw token (for the cookie) and the CSRF token.
`readSession(db: Db, raw: string)`	Looks up a valid, unexpired session for an active user by raw token; null if none.
`deleteSession(db: Db, raw: string, tenantId: string)`	Deletes the session (logout).
`safeEqual(a: string, b: string)`	Constant-time string comparison (CSRF token check) to avoid timing leaks.

apps/api/src/config.ts

Environment configuration, validated with Zod at start-up (fails fast on missing/invalid values).

`loadConfig(overrides: Record<string, string | undefined> = {})`	Loads .env from the repo root, applies overrides (used by tests) and returns the validated config.

apps/api/src/routes/admin.ts

Administration (brief §4.15): norm-time patterns, holiday calendars, absence codes, employee settings.

`adminRoutes` `const`	Registers the administration endpoints (settings, norm patterns, holidays, absence codes, employee settings).

apps/api/src/routes/api-tokens.ts

API token routes (D-028) — GRASP Controller: Finance/Management/Admin manage their own read-only analytics tokens.

`apiTokenRoutes` `const`	Registers the API token endpoints under /api/v1.

apps/api/src/routes/approval.ts

Approval routes (docs/spec/approval.md) — GRASP Controller: validate input and delegate to `services/approval.ts`, which also checks that the caller is the approver.

`approvalRoutes`	`const`	Registers the approval endpoints under /api/v1.

apps/api/src/routes/calendar.ts

Outlook calendar import routes (docs/spec/integrations.md R6) — GRASP Controller: validate input and delegate to `services/outlook.ts`.

`calendarRoutes`	`const`	Registers the calendar import endpoints under /api/v1.

apps/api/src/routes/data-import.ts

Data import routes (D-024) — GRASP Controller: checks the role and delegates to `services/data-import.ts`.

`dataImportRoutes`	`const`	Registers the import endpoints under /api/v1.

apps/api/src/routes/expenses.ts

Expense and receipt routes (brief §4.7, docs/spec/expenses.md) — GRASP Controller: validate input, delegate to `services/expenses.ts` and `services/files.ts`.

`expenseRoutes`	`const`	Registers the expense and file endpoints under /api/v1.

apps/api/src/routes/finance.ts

Accounting periods, month-end report, price lists and payment plans (brief §4.9, §4.11).

`financeRoutes`	`const`	Registers accounting-period, month-end report, price-list and payment-plan endpoints.

apps/api/src/routes/governance.ts

Audit log and GDPR routes (docs/spec/administration.md, D-025/D-026) — GRASP Controller: check the capability, validate input, delegate to `services/audit-log.ts` and `services/gdpr.ts`.

`governanceRoutes`	`const`	Registers the audit and GDPR endpoints under /api/v1.

apps/api/src/routes/invoicing.ts

Invoicing routes (brief §4.10, docs/spec/invoicing.md) — GRASP Controller.

`invoicingRoutes`	`const`	Registers the invoicing endpoints under /api/v1.

apps/api/src/routes/payroll.ts

Payroll export routes (docs/spec/payroll.md, D-023) — GRASP Controller: Finance only (AC7); validate input and delegate to `services/payroll.ts`.

`payrollRoutes`	`const`	Registers the payroll endpoints under /api/v1.

apps/api/src/routes/planning.ts

Resource planner routes (brief §4.8) — GRASP Controller: validate input, delegate to `services/planning.ts`.

`planningRoutes`	`const`	Registers the resource planner endpoints under /api/v1.

apps/api/src/routes/projects.ts

Project, task and customer routes (docs/spec/projects.md, crm.md) — GRASP Controller: validate input, check the role, delegate to services (`projects.ts` read models, `project-setup.ts` changes, `customers.ts`).

`projectRoutes`	`const`	Registers the project, task and customer endpoints under /api/v1.

apps/api/src/routes/reports.ts

Report routes (docs/spec/reporting.md) — GRASP Controller: validate filters, delegate to `services/reports.ts`, and choose JSON or CSV download.

`reportRoutes` `const`	Registers the report endpoints under /api/v1.

apps/api/src/routes/timesheet.ts

Timesheet routes (docs/spec/time-tracking.md) — GRASP Controller: validate input, check access, delegate to `services/timesheet.ts`.

`timesheetRoutes` `const`	Registers the timesheet endpoints under /api/v1.

apps/api/src/routes/users.ts

People endpoints.

`userRoutes` `const`	Registers /me, /users (scoped by role) and /users/:id/roles (admin, audited).

apps/api/src/server.ts

Process entry point for the API (`pnpm dev` / `pnpm start`): builds the app, listens on API_PORT, runs the outbox delivery loop every 30 s, weekly reminders, a daily retention sweep, and shuts down cleanly on SIGINT/SIGTERM.

Integrationer (packages/integrations)

packages/integrations/src/adapter.ts

Accounting integration adapter (brief §4.14, docs/spec/integrations.md).

`CustomerDoc` `interface`	Accounting integration adapter (brief §4.14, docs/spec/integrations.md).
`IntegrationError` `class`	Error from an external system.

packages/integrations/src/calendar.ts

Calendar adapter (docs/spec/integrations.md R6) — GRASP Polymorphism + Protected Variations: the app only knows `CalendarSource`; Microsoft Graph is one implementation.

`CalendarEvent` `interface`	A calendar event with local wall-clock times in Europe/Copenhagen ("YYYY-MM-DDTHH:mm:ss…").
`CalendarSource` `interface`	Anything that can list a person's events between two dates (from inclusive, to exclusive, "YYYY-MM-DD").
`fromGraphEvent(e: GraphEvent)`	Maps a Graph event to our neutral shape (pure; tested without network).
`GraphCalendarSource` `class`	Reads the signed-in user's calendar through Microsoft Graph with a delegated access token.

packages/integrations/src/economic.ts

e-conomic adapter (REST API, https://restapi.e-conomic.com).

`toEconomicCustomer` `const`	Maps our customer document to the e-conomic customer payload (pure function, tested without network).
`toEconomicDraft` `const`	Amounts in e-conomic are decimal kroner; quantities may be negative on credit notes.
`EconomicAdapter` `class`	AccountingAdapter for e-conomic's REST API.
`economicFromEnv(env: Record<string, string | undefined>)`	Builds the adapter from environment variables, or null when not configured (the default).

Web: komponenter og hjælpere (apps/web/src)

apps/web/src/components/app-shell.tsx

Application frame: top bar (wordmark, language, user, sign out) and the role-filtered main navigation.

`AppShell({ children }: { children: ReactNode })`	Wraps every signed-in page with the top bar, navigation and a skip-to-content link.

apps/web/src/components/dashboard.tsx

Home-page dashboard pieces (pure view components): the role-based to-do list and the key-figure panel, both fed by GET /me/dashboard (apps/api/src/services/dashboard.ts).

`Todo` `interface`	One to-do as returned by the API.
`WeekKpi` `interface`	One complete ISO week of key figures (hours as decimals, percentages in basis points).
`MonthRevenue` `interface`	Invoiced revenue for one month (øre), with the effective hourly rate (øre per hour).
`DashboardData` `interface`	Shape of GET /me/dashboard.
`TodoList({ todos }: { todos: Todo[] })`	The person's to-do list; each line links to where it is solved.
`KpiPanel({ kpis }: { kpis: NonNullable<DashboardData["kpis"]> })`	Key figures for the team or company: utilisation, billable share, a 12-week trend and (Finance) revenue.

apps/web/src/components/expense-approvals.tsx

Expense approval list for line managers and Finance (brief §4.7, decision D-019): each submitted expense with receipt link, approve, or reject with a required comment.

`ExpenseApprovals()`	Lists submitted expenses waiting for the signed-in approver.

apps/web/src/components/gdpr-section.tsx

GDPR actions for one employee (docs/spec/administration.md, D-026), shown to Admin in the employee panel: download everything about the person (data-subject export), and erase a former employee — with a preview of what is kept (bookkeeping, pseudonymised) and removed, blockers explained, a required reason and a confirmation.

`GdprSection({ userId }: { userId: string })`	Export + erasure controls for the employee with `userId`.

apps/web/src/components/language-switch.tsx

DA/EN language toggle.

`LanguageSwitch()`	Two pressed-state buttons (Dansk / English) that switch the UI language immediately.

apps/web/src/components/new-project.tsx

"Nyt projekt" form on the projects page (docs/spec/projects.md).

`NewProject({ onCancel }: { onCancel: ()`	Short form: the essentials only.

apps/web/src/components/payment-plan.tsx

Payment plan card on the project page: fixed-price milestones / retainer fees (brief §4.3, §4.10).

`PaymentPlan({…})`	Fixed-price payment plan / retainer fee schedule on the project page (brief §4.3).

apps/web/src/components/power-bi-access.tsx

"Power BI og Excel" card on the reports page (D-028): the dataset addresses to paste into Power BI / Excel, and the person's own read-only API tokens — create (shown once, with a copy button), see when last used, revoke.

`DATASETS` `const`	The datasets Power BI can load (see apps/analytics/arc_analytics/queries.py).
`PowerBiAccess()`	Card with dataset URLs and the person's Power BI tokens.

apps/web/src/components/progress.tsx

Budget consumption bar: neutral up to 90 %, amber 90–100 %, deep orange over budget (never brand red).

`Progress({ bp, label }: { bp: number | null; label: string })`	Budget consumption bar: neutral up to 90 %, amber 90–100 %, deep orange over budget (never brand red).
`HealthDot({ health, label }: { health: string; label: string })`	Small coloured dot + label for budget health (ok, warning, over, no budget); colour is never the only signal.

apps/web/src/components/project-approvals.tsx

Project manager approval list (docs/spec/approval.md, mode pm_then_line_manager, decision D-022): each employee's submitted hours on the caller's projects, grouped per week and project, with the entries behind them.

`ProjectApprovals()`	Lists the project manager's open approval steps.

apps/web/src/components/session.tsx

Session context for the signed-in user.

`SessionGate({ children, fallback }: { children: ReactNode; fallback: ReactNode })`	Loads the signed-in user; redirects to /login when there is no session.
`useMe()`	Returns the signed-in user (with roles and tenant).

apps/web/src/components/timesheet/calendar-panel.tsx

"From my calendar" panel on the timesheet (docs/spec/integrations.md R6): lists the week's Outlook meetings (read only), lets the employee tick some and register them as draft hours on one task, or ignore them.

`CalendarPanel({ weekStart, onClose }: { weekStart: string; onClose: ()`	The calendar panel for one week; `onClose` hides it.

apps/web/src/components/timesheet/day-view.tsx

Day view (docs/spec/time-tracking.md): one day at a time, hours and comment side by side — handy when writing comments or registering on a phone-sized screen.

`DayView({…})`	Day view of the timesheet: day picker for the week plus one row per task with hours and comment side by side.

apps/web/src/components/timesheet/task-picker.tsx

Accessible combobox for adding a task row (WAI-ARIA combobox pattern): type to search, ↑↓ to move, Enter to add, Esc to close.

`TaskPicker({ onPick, exclude }: { onPick: (t: TaskRef)`	Accessible combobox for adding a task row (WAI-ARIA pattern): type to search, ↑↓ to move, Enter to add, Esc to close.

apps/web/src/components/timesheet/types.ts

Shared types for the timesheet screens, mirroring the GET /api/v1/timesheet/week/:weekStart response.

`TaskRef` `interface`	Shared types for the timesheet screens, mirroring the GET /api/v1/timesheet/week/:weekStart response.
`taskRowKey` `const`	Stable row key for a task row in the timesheet grid.
`absenceRowKey` `const`	Stable row key for an absence-code row in the timesheet grid.

apps/web/src/components/timesheet/week-grid.tsx

Weekly timesheet grid (docs/spec/time-tracking.md).

`rowLabel` `const`	Human-readable label for a timesheet row (used for accessible cell names and panels).
`WeekGrid({ week, rows, selected, onSelect, onCommentNeeded }: Props)`	Keyboard-first weekly timesheet grid.

apps/web/src/lib/api.ts

Thin client for the ARCnordic Tid API.

`ANALYTICS_URL` `const`	Base URL of the Python analytics service (KPI Excel export, Power BI datasets).
`setCsrfToken` `const`	Remembers the session's CSRF token (from login or /me); it is sent on every mutating request.
`ApiError` `class`	Error thrown by `api` for non-2xx responses; `code` is the API's error code (translate with `err.<code>`).
`api(path: string, init: RequestInit = {})`	Calls the ARCnordic Tid API under /api/v1 with the session cookie, JSON body and CSRF header; returns parsed JSON.
`can` `const`	True when the signed-in person may do the action — the single check the web uses (D-025).
`UploadedFile` `interface`	Metadata of an uploaded file (receipt) as returned by POST /files.
`uploadFile(file: File)`	Uploads one file (e.g.

apps/web/src/lib/format.ts

Display formatting for the web app (Danish first): hours, money, percentages, dates and week labels.

`fmtHours(h: number | null | undefined, l: Locale, blankZero = false)`	7.5 → "7,50" (da) / "7.50" (en); 0 → "" when blankZero.
`fmtFlex(h: number, l: Locale)`	Signed hours for flex: +0,50 / −1,25
`fmtKr(minor: number | null | undefined, l: Locale, decimals = 0)`	øre → "117.600 kr" (whole kroner for dashboards)
`fmtPct(bp: number | null | undefined, l: Locale)`	basis points → "49,0 %"
`fmtDay(date: string, l: Locale, opts: Intl.DateTimeFormatOptions)`	Formats a LocalDate with Intl options in the UI locale (UTC-based, so the day never shifts).
`weekLabel(monday: string, l: Locale)`	Label for a week, e.g.

Python-analysetjeneste (apps/analytics)

apps/analytics/arc_analytics/db.py

Database access for the analytics service — GRASP Pure Fabrication.

`load_env()`	Loads KEY=VALUE lines from the repo's .env into os.environ (existing variables win).
`database_url()`	The read-only connection string (ANALYTICS_DATABASE_URL, or the local development default).
`Caller` `class`	The signed-in person: who, in which tenant, with which roles.
`tenant_cursor(tenant_id: str)`	Yields a dict-row cursor in a read-only transaction scoped to one tenant by row-level security.
`caller_from_session(raw_token: str)`	Resolves a raw session cookie to the caller (None when missing, expired or the user is inactive).
`caller_from_api_token(raw_token: str)`	Resolves a personal API token (Power BI / Excel, D-028) to its owner; None when unknown, revoked or expired.

apps/analytics/arc_analytics/excel.py

Excel export — GRASP Pure Fabrication: turns KPI rows into a tidy .xlsx workbook (openpyxl).

`kpi_workbook(projects: list[dict[str, Any]], employees: list[dict[str, Any]], date…)`	Builds the KPI workbook (sheets: Projekter, Medarbejdere, Om) and returns the .xlsx bytes.

apps/analytics/arc_analytics/kpi.py

Key figures — the Python mirror of packages/domain/src/kpi.ts (GRASP Information Expert).

`div_round(numerator: int, denominator: int)`	Integer division rounding half away from zero (mirror of divRound in money.ts).
`utilisation(billable: int, norm: int)`	Utilisation ("udnyttelsesgrad"): billable hours ÷ norm hours, in basis points.
`billable_pct(billable: int, work: int)`	Billable share ("faktureringsgrad"): billable ÷ all registered work hours, in basis points.
`effective_rate(invoiced: int, hours: int)`	Effective hourly rate: invoiced øre ÷ invoiced centi-hours, in øre per hour (None without hours).
`margin(revenue: int, cost: int)`	Contribution margin ("dækningsbidrag") in øre and margin % ("DB-grad") in basis points.

apps/analytics/arc_analytics/main.py

ARCnordic Tid analytics service (D-002, D-021) — FastAPI, read-only.

`health()`	Liveness probe (no database access).
`kpis(caller: CallerDep, p: PeriodDep)`	Project and employee KPIs for the period (hours in centi-hours, money in øre, percentages in basis points).
`kpis_excel(caller: CallerDep, p: PeriodDep)`	The KPI report as an Excel download.
`get_dataset(name: str, caller: CallerDep, p: PeriodDep)`	A flat dataset for Power BI's Web connector or Excel's Power Query.

apps/analytics/arc_analytics/queries.py

Read-only report queries — GRASP Pure Fabrication: SQL that gathers raw numbers; the KPI maths stays in kpi.py.

`project_kpis(cur: psycopg.Cursor, date_from: str, date_to: str)`	Per project: hours, billable hours, cost, invoiced revenue and invoiced hours → billable %, rate, margin.
`employee_kpis(cur: psycopg.Cursor, date_from: str, date_to: str)`	Per employee: work hours, billable hours, absence hours and billable %.
`kpi_dataset(cur: psycopg.Cursor, name: str, date_from: str, date_to: str)`	Project or employee KPIs as a flat table in BI-friendly units: hours and kroner as decimals, shares as fractions.
`dataset(cur: psycopg.Cursor, name: str, date_from: str, date_to: str, sees_re…)`	Rows of one flat dataset (see DATASETS); raises KeyError for an unknown name.

10. API-endpoints

106 endpoints under `/api/v1`. "Adgang" er den kapabilitet (fra rollematrixen) eller regel, ruten tjekker; services begrænser derudover, *hvis* data man ser.

Metode	Sti	Fil	Adgang
GET	/api/v1/admin/settings	routes/admin.ts	settings.read
PATCH	/api/v1/admin/approval-mode	routes/admin.ts	settings.manage
POST	/api/v1/admin/norm-patterns	routes/admin.ts	settings.manage
POST	/api/v1/admin/holiday-calendars/:id/generate	routes/admin.ts	settings.manage
POST	/api/v1/admin/holidays	routes/admin.ts	settings.manage
DELETE	/api/v1/admin/holidays/:id	routes/admin.ts	settings.manage
PATCH	/api/v1/admin/absence-codes/:id	routes/admin.ts	settings.manage
GET	/api/v1/admin/employees/:id	routes/admin.ts	settings.read
PATCH	/api/v1/admin/employees/:id	routes/admin.ts	settings.manage / settings.read
POST	/api/v1/admin/employees/:id/norms	routes/admin.ts	settings.manage
GET	/api/v1/api-tokens	routes/api-tokens.ts	analytics.token
POST	/api/v1/api-tokens	routes/api-tokens.ts	analytics.token
DELETE	/api/v1/api-tokens/:id	routes/api-tokens.ts	analytics.token
GET	/api/v1/approvals/projects	routes/approval.ts	login + regler i service
POST	/api/v1/approvals/projects/decide	routes/approval.ts	login + regler i service
GET	/api/v1/approvals	routes/approval.ts	login + regler i service
POST	/api/v1/approvals/:periodId/approve	routes/approval.ts	login + regler i service
POST	/api/v1/approvals/bulk-approve	routes/approval.ts	login + regler i service
POST	/api/v1/approvals/:periodId/reject	routes/approval.ts	login + regler i service
POST	/api/v1/approvals/:periodId/reopen	routes/approval.ts	login + regler i service
GET	/api/v1/timesheet/calendar/:weekStart	routes/calendar.ts	login + regler i service
POST	/api/v1/timesheet/calendar/import	routes/calendar.ts	login + regler i service
POST	/api/v1/timesheet/calendar/ignore	routes/calendar.ts	login + regler i service
POST	/api/v1/import/:kind/preview	routes/data-import.ts	data.import / people.import
POST	/api/v1/import/:kind/commit	routes/data-import.ts	data.import / people.import
GET	/api/v1/expenses/settings	routes/expenses.ts	login + regler i service
GET	/api/v1/expenses	routes/expenses.ts	login + regler i service
POST	/api/v1/expenses	routes/expenses.ts	login + regler i service
PUT	/api/v1/expenses/:id	routes/expenses.ts	login + regler i service
DELETE	/api/v1/expenses/:id	routes/expenses.ts	login + regler i service
POST	/api/v1/expenses/submit	routes/expenses.ts	login + regler i service
GET	/api/v1/expense-approvals	routes/expenses.ts	login + regler i service

POST	/api/v1/expenses/:id/approve	routes/expenses.ts	login + regler i service
POST	/api/v1/expenses/:id/reject	routes/expenses.ts	login + regler i service
POST	/api/v1/files	routes/expenses.ts	login + regler i service
GET	/api/v1/files/:id	routes/expenses.ts	login + regler i service
GET	/api/v1/periods	routes/finance.ts	period.view
GET	/api/v1/periods/:month/report	routes/finance.ts	period.view
POST	/api/v1/periods/:id/close	routes/finance.ts	period.close
POST	/api/v1/periods/:id/reopen	routes/finance.ts	period.close
GET	/api/v1/price-lists	routes/finance.ts	login + regler i service
POST	/api/v1/price-lists/:id/rates	routes/finance.ts	pricelist.manage
DELETE	/api/v1/price-list-rates/:id	routes/finance.ts	pricelist.manage
GET	/api/v1/projects/:id/payment-plan	routes/finance.ts	login + regler i service
POST	/api/v1/contracts/:id/plan-items	routes/finance.ts	login + regler i service
PATCH	/api/v1/plan-items/:id	routes/finance.ts	login + regler i service
GET	/api/v1/audit	routes/governance.ts	audit.read
GET	/api/v1/audit/entities	routes/governance.ts	audit.read
GET	/api/v1/admin/employees/:id/gdpr-export	routes/governance.ts	gdpr.manage
GET	/api/v1/admin/employees/:id/erasure	routes/governance.ts	gdpr.manage
POST	/api/v1/admin/employees/:id/erase	routes/governance.ts	gdpr.manage
GET	/api/v1/admin/retention	routes/governance.ts	gdpr.manage
GET	/api/v1/invoicing/basis	routes/invoicing.ts	login + regler i service
POST	/api/v1/invoices/drafts	routes/invoicing.ts	login + regler i service
POST	/api/v1/invoices/bulk-drafts	routes/invoicing.ts	invoice.prepare_all
GET	/api/v1/invoices	routes/invoicing.ts	login + regler i service
GET	/api/v1/invoices/:id	routes/invoicing.ts	login + regler i service
PATCH	/api/v1/invoices/:id	routes/invoicing.ts	login + regler i service
PATCH	/api/v1/invoice-lines/:id	routes/invoicing.ts	login + regler i service
POST	/api/v1/invoices/:id/lines	routes/invoicing.ts	login + regler i service
DELETE	/api/v1/invoice-lines/:id	routes/invoicing.ts	login + regler i service
DELETE	/api/v1/invoices/:id	routes/invoicing.ts	login + regler i service
POST	/api/v1/invoices/:id/ready	routes/invoicing.ts	login + regler i service
POST	/api/v1/invoices/:id/book	routes/invoicing.ts	invoice.book
POST	/api/v1/invoices/:id/credit	routes/invoicing.ts	invoice.credit
GET	/api/v1/invoices/:id/html	routes/invoicing.ts	login + regler i service
GET	/api/v1/invoices/:id/pdf	routes/invoicing.ts	login + regler i service
GET	/api/v1/payroll/periods	routes/payroll.ts	payroll.export
GET	/api/v1/payroll/periods/:month	routes/payroll.ts	payroll.export

POST	/api/v1/payroll/periods/:month/export	routes/payroll.ts	payroll.export
POST	/api/v1/payroll/periods/:month/lock	routes/payroll.ts	payroll.export
GET	/api/v1/payroll/periods/:month/file.csv	routes/payroll.ts	payroll.export
GET	/api/v1/payroll/salary-codes	routes/payroll.ts	payroll.setup
PATCH	/api/v1/payroll/salary-codes/:id	routes/payroll.ts	payroll.setup
PUT	/api/v1/payroll/absence-mapping	routes/payroll.ts	payroll.setup
GET	/api/v1/planning	routes/planning.ts	login + regler i service
PUT	/api/v1/planning/allocations	routes/planning.ts	login + regler i service
GET	/api/v1/planning/competencies	routes/planning.ts	login + regler i service
PUT	/api/v1/users/:id/competencies	routes/planning.ts	login + regler i service
GET	/api/v1/projects	routes/projects.ts	login + regler i service
GET	/api/v1/projects/:id	routes/projects.ts	login + regler i service
POST	/api/v1/projects	routes/projects.ts	project.create
POST	/api/v1/projects/:id/tasks	routes/projects.ts	login + regler i service
PATCH	/api/v1/tasks/:id	routes/projects.ts	login + regler i service
GET	/api/v1/customers	routes/projects.ts	login + regler i service
POST	/api/v1/customers	routes/projects.ts	login + regler i service
GET	/api/v1/me/balances	routes/reports.ts	login + regler i service
GET	/api/v1/me/dashboard	routes/reports.ts	login + regler i service
GET	/api/v1/reports/budget-actual	routes/reports.ts	login + regler i service
GET	/api/v1/reports/time-entries	routes/reports.ts	login + regler i service
GET	/api/v1/timesheet/week/:weekStart	routes/timesheet.ts	login + regler i service
PUT	/api/v1/timesheet/entries	routes/timesheet.ts	login + regler i service
POST	/api/v1/timesheet/week/:weekStart/submit	routes/timesheet.ts	login + regler i service
POST	/api/v1/timesheet/week/:weekStart/reopen	routes/timesheet.ts	login + regler i service
GET	/api/v1/timesheet/tasks	routes/timesheet.ts	login + regler i service
POST	/api/v1/timesheet/favourites	routes/timesheet.ts	login + regler i service
DELETE	/api/v1/timesheet/favourites/:taskId	routes/timesheet.ts	login + regler i service
GET	/api/v1/timesheet/week/:weekStart/previous-rows	routes/timesheet.ts	login + regler i service
GET	/api/v1/me	routes/users.ts	login + regler i service
GET	/api/v1/users	routes/users.ts	login + regler i service
PUT	/api/v1/users/:id/roles	routes/users.ts	settings.manage
GET	/api/v1/auth/providers	auth/routes.ts	offentlig
POST	/api/v1/auth/dev-login	auth/routes.ts	offentlig
POST	/api/v1/auth/logout	auth/routes.ts	login + regler i service
GET	/api/v1/auth/entra/login	auth/routes.ts	offentlig
GET	/api/v1/auth/entra/callback	auth/routes.ts	offentlig

11. Datamodel

51 tabeller i PostgreSQL (`packages/db/src/schema.ts`). Alle tabeller med `tenant_id` er beskyttet af row-level security.

Tabel	Kolonner	Formål
`tenants`	9	
`legal_entities`	8	
`departments`	9	
`users`	22	
`user_roles`	3	
`sessions`	6	
`audit_log`	10	
`norm_patterns`	6	
`employee_norms`	7	
`holiday_calendars`	4	
`holidays`	7	
`absence_codes`	11	
`customers`	16	
`contacts`	7	
`projects`	17	
`contracts`	10	
`tasks`	16	
`project_members`	3	
`timesheet_periods`	10	
`timesheet_events`	7	
`time_entries`	21	
`favourites`	3	
`price_groups`	4	Rate role of an employee (e.g.
`task_types`	3	
`price_lists`	7	
`price_list_rates`	7	
`project_employee_rates`	6	
`payment_plan_items`	8	Fixed-price payment plan / retainer fee schedule.
`number_series`	5	Gap-free invoice numbering per legal entity (brief §4.10).
`invoices`	27	
`invoice_lines`	16	
`invoice_line_sources`	7	Which time entries an invoice line covers, with registered and invoiced value (write-down analytics).

accounting_periods	7	
revenue_recognitions	8	Snapshot taken when a month is closed: fixed-price revenue recognised to date and the month's delta.
integration_connections	7	Configured accounting/payroll connections.
outbox	9	Transactional outbox: events written in the same transaction as the change, delivered by a worker.
expense_types	6	Expense categories; `vatDeductibleBp` is the share of the VAT the company may deduct (e.g.
mileage_rates	6	Mileage allowance per year: high rate (øre/km) up to the threshold, low rate above.
files	8	Uploaded files (receipts).
expenses	26	An out-of-pocket expense or a mileage claim.
allocations	7	Planned hours: one employee on one project in one ISO week (decision OQ-28 default: weekly granularity).
user_competencies	3	Competencies as simple tags per employee (e.g.
reminder_log	6	E-mail reminders already sent, so the weekly job never sends the same reminder twice.
graph_tokens	4	Delegated Microsoft Graph access token per user (read-only scope Calendars.Read), AES-GCM encrypted with a key derived from SESSION_SECRET.
calendar_imports	9	Calendar events (per day) the user has imported into the timesheet or chosen to ignore — shown as handled.
salary_codes	9	Salary codes ("lønarter") the payroll system knows.
pay_periods	8	Monthly pay periods: open → exported → locked.
payroll_runs	7	One generated export file for a pay period.
payroll_lines	9	Lines of a run; `month` is the month the items are dated in (earlier than the run's month for late lines).
user_rights	4	Extra rights on top of roles (D-025), e.g.
api_tokens	11	Personal read-only tokens for the analytics service (spec integrations R10, decision D-028), e.g.

12. Beslutningslog

ID	Dato	Beslutning	Status
D-001	2026-11-08	Repository location	accepted
D-002	2026-11-08	Languages: TypeScript core, Python analytics service	accepted
D-003	2026-11-08	Contracts are child objects of projects	accepted
D-004	2026-11-08	Fixed-price completion	accepted
D-005	2026-11-08	Hour input formats	accepted
D-006	2026-11-08	Approval granularity	accepted
D-007	2026-11-08	Visual identity	proposed
D-008	2026-11-08	Retainers as a contract type	accepted
D-009	2026-11-08	Toolchain	accepted
D-010	2026-11-08	Product name in code	accepted
D-011	2026-11-08	Backend stack details	accepted
D-012	2026-11-08	Multi-tenancy enforcement	accepted
D-013	2026-11-08	Business-rule defaults (owner delegated: "just make something")	
D-014	2026-11-08	Contract hourly rate in the rate order	accepted
D-015	2026-11-08	Invoice numbering and immutability	accepted
D-016	2026-11-08	PDF rendering	accepted
D-017	2026-11-08	Run locally, no hosting for now	accepted
D-018	2026-11-08	Code structure (GRASP) and documentation rules	accepted
D-019	2026-11-08	Expenses: approval, VAT, mileage rates, receipts	accepted
D-020	2026-11-08	Dashboard, reminders and Outlook calendar import	accepted
D-021	2026-11-08	Python analytics service	accepted
D-022	2026-11-08	Approval modes	accepted
D-023	2026-11-08	Payroll export, first version	accepted
D-024	2026-11-08	Data import	accepted
D-025	2026-11-09	Access policy and segregation of duties	accepted
D-026	2026-11-09	GDPR	accepted
D-027	2026-11-09	Operations	proposed
D-028	2026-11-09	Power BI / Excel access with personal tokens	accepted

Fuld tekst med begrundelser: `docs/decisions.md`.

13. Kendte begrænsninger og anbefalinger

Kræver beslutning hos ejeren

- **Hosting (OQ-04)** — forslag i `docs/operations.md`; indtil da kun lokal test med opdigtede data.
- **Flyt arbejdskopien ud af OneDrive (OQ-01)** — trin-for-trin i `docs/operations.md` §4.
- **Åbne fagspørgsmål** — flexloft, ferieoptjening, kilometersatser og moms (OQ-36), nummerserier, lønudbyder (OQ-08), offentlige kunder/OIOUBL (OQ-32), opbevaringsfrist og regnskabsår (OQ-39). Systemet kører på standardvalg indtil da.

Bevidste forenklinger

- Løneksport er generisk CSV; adaptere til Danløn/DataLøn/Zenegy kommer, når udbyderen er valgt.
- Persondata-udtræk er én JSON-fil (spec nævner ZIP med JSON+CSV).
- Revisionssporet er append-only men endnu ikke hash-kædet (spec R2 foreslår kæde for manipulationssikring).
- Kalender-tokens fornyes ikke automatisk (ingen refresh-tokens gemmes) — brugeren logger ind med Microsoft igen.
- Bogføringsdata ud over 5 år rapporteres, men slettes ikke automatisk.

Gode næste skridt

- Entra ID med automatisk oprettelse og gruppe → rolle (OQ-10), så roller ikke vedligeholdes to steder.
- Projektskabeloner med faser fra ydelsesbeskrivelsen (OQ-23).
- Mobil/PWA til tidsregistrering på byggepladsen.
- API-tokens (spec R10), så Power BI kan opdatere automatisk.